C201/C255/I211/A505 Homework #1 Due 2/5/26

This homework deals with writing a "censoring" program that works as follows:

1) The program repeatedly prompts the user to enter a sentence to be censored.
2) After a sentence is entered, the program searches for the word "hell" and replaces it with "heck".
3) The original and censored sentences are displayed within quotes.
4) The program should continue to ask for sentences to censor until user enters nothing, enter key.
5) Uses a try-catch block to check for array out of bounds.

Output should look like following:

```
Enter Sentence ==> Hell, its hot as hell looking for shells in hell!

Original : "Hell, its hot as hell looking for shells in hell!"
Censored : "Heck, its hot as heck looking for shells in heck!"
-------------------------------------------------------------------

Enter Sentence ==> Hello Hell.

Original : "Hello Hell."
Censored : "Hello Heck."
-------------------------------------------------------------------

Enter Sentence ==>
Goodbye.....
```

<u>Hints and Suggestions:</u>

1) Write program so that output looks like the above. Following deductions will be made:
   - 4 points if program does not censor sentence containing a mixture of upper and lower case letters.
   - 3 points if program displays the censored sentence in all uppercase letters (or lowercase).
   - 3 points if program does not handle punctuation.
   - 3 points if quotes not displayed around outputted original and censored sentence.
   - 2 points if separating lines ( ---------- ) not displayed
   - 2 points if you make “, quotes, part of the string.

2) Think carefully about the design of the functions that you write for this program, these should be logically coherent. Your `main()` should not contain any messy detail. Remember many points depend on program design and organization.

3) The algorithm for this program is non-trivial. Be sure to spend sufficient time constructing the algorithm, and do not rush right into coding. We will go over basic algorithm in class.

4) Think carefully about the test data for this program. It would be quite easy to miss logical errors unless data is well chosen.

5) This assignment is designed to be done with character arrays, and the algorithm we cover will use character arrays. If you choose to use Strings, you may find it more difficult to do.

6) To test for the try-catch, change array size to something small, like 5 or 6.
Sample output for this catch:

```
Enter sentence==> helloss
An Array went out of bounds
Program terminated
```

## How to submit your homework's (For online students only)

All homework's are due by the posted date on the top of assignment and by midnight. If turned in one class period late then -3 points and -6 points if turned in two class periods late by midnight, nothing will be accepted after that time. Everything will be submitted via email, do NOT use canvas or canvas messages to send me anything. Use your school or personal email only and send to my school email or personal email given on the syllabus.

Send the completed homework as an attachment with one and only one pdf file. Make sure you have your name on the assignment as well.

**IMPORTANT**: Make sure to submit one pdf file that contains all the programs and or algorithms. Each homework problem will have its own program file with output that will need to be saved as a pdf file. All the separate pdf files will then need to be combined into one pdf file to send to me. Make sure the combined pdf files are in order as given in homework page.

If you have a handwritten algorithm to submit, then there are two ways listed below to save file as pdf.

1) You can scan the document on a printer and save as a pdf.
2) You can download an app on your phone that allows you to take picture of document and it will then save it as a pdf file.

Combining pdf files several ways, choose a method you know how, or try on listed below.

1) Go to https://iuware.iu.edu/ and download the "Creative Cloud" from adobe, then can install "Acrobat Pro" for dealing with pdf files. ( I called OIT and had them help with this).
2) Can try LightPDF, https://lightpdf.com/, I have not used this but other students in past have used it.

Even if you are unable to complete the assignment, turn what you have done in. Something is worth some points, nothing is a zero.